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Assessment tool-2
CO2
Scenario-based

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1. Functional Blocks of a machine Learning System

Machine Learning (ML) systems learns from data and makes predictions or decisions

Data collection
↓
Data preprocessing
↓
Feature extraction
↓
model training
↓
model evaluation
↓
prediction/output

* Collect data from sensors, databases, images, or websites
* Remove errors, handle missing values, and normalize data
* Select important information needed for prediction
* Train the algorithm using the collected data
* check the accuracy and performance of the model
* use the trained model to predict new results.

2. Machine Learning System for disaster management

Aim: TO predict disaster such as floods and earthquakes using historical and real-time data

Block diagram:

Historical Data + Real-time Sensor

↓

. Data preprocesing

↓

Feature selection

↓

ML model traning and testing

* Collect rainfall, river-water levels, weather reports, seismic readings, and past disaster records.
* Remove incorrect data, fill missing values, remove duplicates and normalize.
* Select important feature such as rain fall, river level, soil moisture, and seismic
* Learns complex pattern from large datasets
* Easy to understand and classify disaster risk
* Useful for predicting changes in rainfall and river
* ML helps authorities assess disaster risks and improve emergency response.

3. Back propagation Algorithm in Neural network

Back propagation is an algorithm used to reduce prediction errors by updating the weights of a neural network

Input layer
↓
Hidden layer
↓
output layer
↓
Calculate error
↓
update weights Backward

* Input data passes through the network to produce an output
* compare the predicted output with the actual output
* Send the error backward through the network to calculate gradients
* update weights to reduce the error
* continue training until the error becomes sufficiently small

Error = 1 - 0.7
= 0.3

$$w_{new} = w_{old} - \eta \frac{\partial \epsilon}{\partial w}$$

w = weight
η = learning rate
ϵ = Error

4. Genetic Algorithm for Speed and Accuracy optimization

A Genetic Algorithm finds a good by repeatedly solution improving a population of candidate solution

Given dataset

Example	Speed	Accuracy	Fitness	Suitable
1	High	Low	40	No
2	Medium	Medium	60	Yes
3	High	Medium	75	Yes
4	Low	High	65	Yes
5	Medium	High	80	Yes

* Start with the five candidate solutions given in the dataset
* Select individuals with higher fitness values because they are better Solution
* Combine selected parents to create new solutions For example.
* Randomly change a parameter to explore new Solutions. For example change speed from medium to High.
* Calculate the fitness of each new solution and retain the better candidates.

5. Neural Network Classification and performance Evaluation

A neural network can predict whether a patient has a disease using age, blood pressure, glucose level, and life style risk.

Working steps:

1. Collect patient data and actual disease labels
2. preprocess the data by handling missing values and converting categories into numerical values
3. Select important features such a blood pressure and glucose level
4. Train the neural Network using forward propagation and back propagation.

* True positive : Actual yes, predicted yes = 2
* True Negative: Actual No, predicted No = 1
* False positive : Actual No, predicted Yes = 1
* False Negative : Actual, yes predicted No = 1

1. $\frac{TP+TN}{TP+TN+FP+FN} \times 100$

$= \frac{2+1}{5} \times 100$

$= 60\%$

2. $\frac{TP}{TP+FP} = \frac{2}{3}$

$= 66.67\%$

3. $\frac{TP}{TP+FP} = \frac{2}{3}$

66.67%

4. $F1 = \frac{2PR}{P+R}$

$= 66.67\%$

* The NN achives 60% accuracy on the given five examples more representative training and test data are required to judge.